

A Counterexample to the Unqualified Semistrict-Graphical-Derivative Identity for Convex C^2 -Decomposable Functions

Abstract

This note gives a counterexample, and therefore a negative solution, to the unqualified question whether every convex C^2 -decomposable function g satisfies the semistrict graphical derivative identity

$$D^b(\partial g)(\bar{u}, \bar{y})(w) = \text{cl}(D(\partial g)(\bar{u}, \bar{y})(w) - K_g(\bar{u}, \bar{y})^*) \quad \text{for all } w.$$

The example is the indicator of the epigraph of $s \mapsto (3/4)|s|^{4/3}$. The function is proper, lower semicontinuous, convex, and C^2 -decomposable, but the displayed identity fails at $(\bar{u}, \bar{y}) = ((0, 0), (0, 0))$ in the direction $w = (0, 0)$. Thus the result is not merely partial progress for the unqualified formulation. It does not address a strengthened version in which the C^2 -decomposition is required to satisfy an additional nondegeneracy or cone-reducibility hypothesis.

1 Introduction and status of the result

Hang and Sarabi study semistrict graphical derivatives in their convergence analysis of augmented Lagrangian methods for composite optimization problems; see [1]. One relevant identity, denoted as equation (3.14) in that paper, is

$$D^b(\partial g)(\bar{u}, \bar{y})(w) = \text{cl}(D(\partial g)(\bar{u}, \bar{y})(w) - K_g(\bar{u}, \bar{y})^*) \quad \text{for all } w. \tag{1}$$

Here D denotes the ordinary graphical derivative, D^b denotes the semistrict graphical derivative, and $K_g(\bar{u}, \bar{y})^*$ is the polar of the critical cone. The unqualified open-problem formulation asks whether (1) follows from convexity and C^2 -decomposability of g alone; see also [2].

The answer is no. The example below is elementary but genuinely C^2 -decomposable. Hence it gives a full negative solution to the unqualified problem. The example is degenerate, so it should not be interpreted as a counterexample to results that include an explicit nondegeneracy assumption, such as the nondegenerate chain-rule result proved in Proposition 3.5 of [1].

2 Definitions and conventions

All spaces in this note are finite-dimensional Euclidean spaces. For a closed convex set $C \subset \mathbb{R}^n$, δ_C denotes its indicator function and

$$N_C(u) := \{y \in \mathbb{R}^n \mid \langle y, z - u \rangle \leq 0 \text{ for every } z \in C\}$$

denotes the convex normal cone. Thus $\partial \delta_C = N_C$.

For a set-valued mapping $F : X \rightrightarrows Y$ and $(\bar{x}, \bar{y}) \in \text{gph } F$, the graphical derivative is defined by

$$\text{gph } DF(\bar{x}, \bar{y}) = T_{\text{gph } F}(\bar{x}, \bar{y}).$$

Equivalently,

$$\eta \in DF(\bar{x}, \bar{y})(w) \iff \exists t_k \downarrow 0, w_k \rightarrow w, \eta_k \rightarrow \eta \text{ such that } \bar{y} + t_k \eta_k \in F(\bar{x} + t_k w_k).$$

The semistrict graphical derivative is the set-valued mapping $D^b F(\bar{x}, \bar{y})$ characterized by

$$\begin{aligned} \eta \in D^b F(\bar{x}, \bar{y})(w) \iff & \exists t_k \downarrow 0, (x_k, y_k) \rightarrow (\bar{x}, \bar{y}), \hat{y}_k \rightarrow \bar{y}, \\ & y_k \in F(x_k), \quad \hat{y}_k \in F(\bar{x}), \quad \frac{x_k - \bar{x}}{t_k} \rightarrow w, \quad \frac{y_k - \hat{y}_k}{t_k} \rightarrow \eta. \end{aligned}$$

This is the semistrict derivative notion used in [1], with the notation D^b used here.

For a set A , we use the polar-cone convention

$$A^* := \{p \mid \langle p, a \rangle \leq 0 \text{ for every } a \in A\}.$$

For a proper lower semicontinuous convex function g and $\bar{y} \in \partial g(\bar{u})$, the critical cone is

$$K_g(\bar{u}, \bar{y}) := \{w \mid dg(\bar{u})(w) = \langle \bar{y}, w \rangle\}.$$

When $g = \delta_C$ and C is closed and convex, this reduces to

$$K_g(\bar{u}, \bar{y}) = T_C(\bar{u}) \cap [\bar{y}]^\perp.$$

We use the following form of C^2 -decomposability. A proper lower semicontinuous convex function g is C^2 -decomposable at $\bar{u}$ if, in a neighborhood of $\bar{u}$,

$$g(u) = g(\bar{u}) + \vartheta(\Xi(u)),$$

where Ξ is a C^2 mapping with $\Xi(\bar{u}) = 0$ and ϑ is proper, lower semicontinuous, convex, and sublinear.

3 The example is convex and C^2 -decomposable

Define

$$f(s) := \frac{3}{4}|s|^{4/3}, \quad C := \text{epi } f = \{(s, r) \in \mathbb{R}^2 \mid r \geq \frac{3}{4}|s|^{4/3}\},$$

and set

$$g := \delta_C, \quad \bar{u} := (0, 0), \quad \bar{y} := (0, 0).$$

Since f is convex, C is closed and convex. Therefore g is proper, lower semicontinuous, and convex, and $0 \in \partial g(\bar{u})$.

Lemma 1. *The function $g = \delta_C$ is C^2 -decomposable at $\bar{u} = (0, 0)$.*

Proof. Let

$$Q := \{(a, b, c) \in \mathbb{R}^3 \mid a^2 \leq bc, b \geq 0, c \geq 0\}.$$

The set Q is the closed convex rotated second-order cone. Thus $\vartheta := \delta_Q$ is proper, lower semicontinuous, convex, and sublinear. Define

$$\Xi(s, r) := ((3/4)^{3/2}s^2, r, r^2).$$

Then Ξ is C^2 and $\Xi(0,0) = 0$. Moreover,

$$\begin{aligned}\Xi(s,r) \in Q &\iff r \geq 0 \text{ and } (3/4)^3 s^4 \leq r^3 \\ &\iff r \geq \frac{3}{4}|s|^{4/3}.\end{aligned}$$

Hence $C = \Xi^{-1}(Q)$, and globally

$$g(s,r) = \delta_C(s,r) = \delta_Q(\Xi(s,r)) = g(\bar{u}) + \vartheta(\Xi(s,r)).$$

This is the desired C^2 -decomposition. $\square$

4 Normal cones, the critical cone, and the ordinary graphical derivative

The derivative of f is

$$f'(s) = \text{sgn}(s)|s|^{1/3} \quad (s \neq 0), \quad f'(0) = 0.$$

For every boundary point $(s, f(s))$ of C , the epigraph normal cone formula gives

$$N_C(s, f(s)) = \{\lambda(f'(s), -1) \mid \lambda \geq 0\}. \quad (2)$$

For interior points of C , the normal cone is $\{0\}$. In particular,

$$N_C(0,0) = \{0\} \times \mathbb{R}_-. \quad (3)$$

Because $\bar{y} = 0$, the critical cone equals $T_C(0,0)$. We compute it explicitly:

$$K_g(\bar{u}, \bar{y}) = T_C(0,0) = \{(h,k) \in \mathbb{R}^2 \mid k \geq 0\}. \quad (4)$$

Indeed, if $(s_j, r_j) \in C$, $t_j \downarrow 0$, and $(s_j, r_j)/t_j \rightarrow (h,k)$, then $r_j \geq 0$, so $k \geq 0$. Conversely, if $k > 0$, then $(th, tk) \in C$ for all small $t > 0$ because tk dominates $(3/4)t^{4/3}|h|^{4/3}$; if $k = 0$, then $(th, f(th))/t \rightarrow (h,0)$. Taking the polar of (4) gives

$$K_g(\bar{u}, \bar{y})^* = \{0\} \times \mathbb{R}_-. \quad (5)$$

Lemma 2. For $w = (0,0)$,

$$D(\partial g)(\bar{u}, \bar{y})(w) = DN_C((0,0), (0,0))(0,0) = \{0\} \times \mathbb{R}_-.$$

Proof. Let $\eta = (\eta_1, \eta_2) \in DN_C((0,0), (0,0))(0,0)$. Then there exist $t_j \downarrow 0$, $u_j = (s_j, r_j) \in C$, and $v_j \in N_C(u_j)$ such that

$$\frac{u_j}{t_j} \rightarrow (0,0), \quad \frac{v_j}{t_j} \rightarrow \eta.$$

For each j , either u_j is an interior point of C , in which case $v_j = 0$; or u_j is a boundary point, in which case (2) gives a multiplier $\lambda_j \geq 0$ such that

$$v_j = \lambda_j(f'(s_j), -1).$$

The same displayed formula is also valid in the interior case by taking $\lambda_j = 0$. Hence $(v_j)_2 \leq 0$ for every j , so $\eta_2 \leq 0$. Moreover, v_j/t_j converges, so $\lambda_j/t_j = -(v_j)_2/t_j$ is bounded. Since $u_j/t_j \rightarrow 0$ and $t_j \downarrow 0$, we have $s_j \rightarrow 0$, and therefore

$$\frac{(v_j)_1}{t_j} = \frac{\lambda_j}{t_j} f'(s_j) \rightarrow 0.$$

Thus $\eta_1 = 0$ and $\eta_2 \leq 0$.

Conversely, for any $\beta \leq 0$, choose $u_j = (0, 0)$ and $v_j = t_j(0, \beta)$. By (3), $v_j \in N_C(0, 0)$, and hence $(0, \beta) \in DN_C((0, 0), (0, 0))(0, 0)$. This proves the claim. $\square$

Combining Lemma 2 with (5), we obtain

$$\begin{aligned} \text{cl}(D(\partial g)(\bar{u}, \bar{y})(0, 0) - K_g(\bar{u}, \bar{y})^*) &= (\{0\} \times \mathbb{R}_-) - (\{0\} \times \mathbb{R}_-) \\ &= \{0\} \times \mathbb{R}. \end{aligned} \tag{6}$$

5 A semistrict derivative vector outside the right-hand side

Lemma 3. *The vector $(1, 0)$ belongs to $D^b(\partial g)(\bar{u}, \bar{y})(0, 0)$.*

Proof. Let $t > 0$ and set

$$s_t := t^2, \quad u_t := (s_t, f(s_t)) = \left(t^2, \frac{3}{4}t^{8/3}\right), \quad \lambda_t := t^{1/3}.$$

Then

$$\frac{u_t}{t} = \left(t, \frac{3}{4}t^{5/3}\right) \rightarrow (0, 0).$$

Since u_t is a boundary point of C and $f'(s_t) = t^{2/3}$, formula (2) gives

$$y_t := \lambda_t(t^{2/3}, -1) = (t, -t^{1/3}) \in N_C(u_t).$$

On the other hand, by (3),

$$\hat{y}_t := (0, -t^{1/3}) \in N_C(0, 0).$$

Both y_t and $\hat{y}_t$ converge to $(0, 0)$, and

$$\frac{y_t - \hat{y}_t}{t} = (1, 0).$$

The defining sequential criterion for the semistrict graphical derivative therefore gives

$$(1, 0) \in D^b N_C((0, 0), (0, 0))(0, 0) = D^b(\partial g)(\bar{u}, \bar{y})(0, 0).$$

$\square$

6 Failure of the identity

Theorem 1. *There exists a proper lower semicontinuous convex C^2 -decomposable function g for which the identity (1) fails. Consequently, convexity and C^2 -decomposability alone do not imply (1).*

Proof. Take the function $g = \delta_C$ constructed above. By Lemma 1, g is convex and C^2 -decomposable. For $w = (0, 0)$, the right-hand side of (1) is $\{0\} \times \mathbb{R}$ by (6). However, Lemma 3 shows that the left-hand side contains $(1, 0)$, and

$$(1, 0) \notin \{0\} \times \mathbb{R}.$$

Thus (1) fails. $\square$

Remark 1 (Scope of the counterexample). *The counterexample refutes the unqualified formulation only. It does not contradict a theorem that assumes an additional nondegeneracy condition on a particular C^2 -decomposition. For the decomposition used in Lemma 1,*

$$\nabla\Xi(0,0)(h,k) = (0,k,0),$$

so the derivative of the reducing map is far from surjective. Thus the example should be interpreted as a negative solution to the unqualified “every C^2 -decomposable function” question, while stronger nondegenerate variants are not addressed by this construction.

References

- [1] N. T. V. Hang and E. Sarabi, *Convergence of augmented Lagrangian methods for composite optimization problems*, Mathematics of Operations Research **51** (2026), no. 1, 591–620. DOI: 10.1287/moor.2023.0324; arXiv: 2307.15627.
- [2] P. Nuti, E. Fithian, and R. Niazadeh, maintainers, *Open Problems in Operations Research*, problem W4408446556_p0, https://pranav-nuti.github.io/open-problems-in-or/problem.html?id=W4408446556_p0&page=13&n=146, accessed June 24, 2026.